

Alemayehu Mekonen

Software Engineer — Backend, Platform & AI

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SUMMARY

Full-stack and backend engineer with production experience across telecom fintech, banking infrastructure and AI-product engineering. Currently split between a senior backend role on Safaricom Ethiopia's M-PESA and USSD mobile-money systems, and a remote AI-product engineering role for a US company. Comfortable owning a service end to end — design, containerisation, deployment and the operational reality of keeping it up — and increasingly building the retrieval and agent systems that sit on top of that data rather than just calling a model API.

EXPERIENCE

Senior Backend Engineer — Safaricom Ethiopia

Aug 2025 – Present · Addis Ababa

- Build and maintain backend microservices for M-PESA LiteApp — secure merchant payments, transaction orchestration, settlements and retry workflows for high-volume financial services.
- Develop backend services for USSD-based mobile money — balance checks, transfers and merchant payments — with reliable fault recovery for low-bandwidth telecom users.
- Stack: Microservices, Spring Boot, Redis, Kubernetes, PostgreSQL, REST/SOAP.

Full Stack Developer — Universal Perk

Jun 2025 – Present · Remote, Texas USA

- Built and launched Creva AI, Dasguzo and other SaaS products, owning frontend, backend, authentication and cloud deployment end to end.
- Developed scalable business platforms and integrations with secure APIs, caching and workflow automation for production delivery.
- Stack: React, Next.js, Node.js, MongoDB, Redis, REST APIs.

DevOps Engineer — Lion International Bank

Jan 2025 – Sep 2025 · Addis Ababa

- Built and maintained secure CI/CD pipelines, containerised services and Kubernetes deployment workflows, improving release reliability and rollback safety for banking applications.
- Developed monitoring, logging, backup and fault-recovery systems for critical banking services, targeting high availability and fast incident response.
- Stack: Kubernetes, Docker, CI/CD, Linux, PostgreSQL, Prometheus.

Full Stack Developer — Addis Software P.L.C

Sep 2024 – Jan 2025 · Addis Ababa

- Collaborated on the Dodai Admin Panel — backend services, production APIs and IoT integrations for real-time device monitoring and telemetry.
- Stack: React, Node.js, MongoDB.

Backend Development Apprentice — Safaricom Ethiopia

Mar 2024 – May 2024 · Addis Ababa

- Two-month certified apprenticeship in backend development — the stint that led to the senior role held there now.

PROJECTS

Authentication & Authorization Service — Spring Boot, PostgreSQL, Redis

Secure authentication service handling user login, JWT token management, role-based access control and session caching.

Reusable React UI Component Library — React, TypeScript, Tailwind CSS

Shared component library with consistent design patterns to accelerate frontend development across multiple projects.

University Management System — React, Node.js, MongoDB

System for managing student academic records, enabling instructors to handle grading and streamline workflows across departments.

Zemenawi CRM & this portfolio's AI assistant — Spring Boot / Spring Cloud · Next.js / Gemini / pgvector

A CRM decomposed into independently deployable microservices (gateway, discovery, auth, config, notifications, logging), and a retrieval-augmented AI agent with tool-calling, source citations and guardrails against fabrication — see project details at the portfolio link above.

SKILLS

Backend & Platform

Java, Spring Boot, Node.js, Microservices, REST APIs, System Design, Fintech Payments, Docker, Kubernetes, Redis, PostgreSQL, MongoDB, Linux, CI/CD, Jenkins

Identity & Access

Keycloak, OAuth2, JWT, Role-Based Access Control

AI Engineering

Retrieval-Augmented Generation (RAG), Prompt Engineering, Tool/Function Calling, Vector Databases, LLM Application Development, AI Agent Development, Agentic Workflows, AI Orchestration

Frontend

TypeScript, React, Next.js, Tailwind CSS

EDUCATION & CERTIFICATES

BSc, Computer Science — Unity University, Adama 2018 – 2023

Certificate of Completion, Back-End Engineering (180 hrs) — Gebeya 2023 – 2024

Certificate of Apprenticeship, Back-End Development — Safaricom Ethiopia 2024

Backend Engineer in Spring Boot Microservices — LinkedIn Learning

Docker & Kubernetes Essentials — Udemy